Name: Student ID:

The code below builds a scatter plot using some of the variables in the `penguins` data frame. Copy this into a new code chunk in a new file, as you'll be modifying it shortly.

```
ggplot(penguins, aes(x = bill_length_mm,
                     y = bill_depth_mm,
                     color = species)) +
    geom_point() +
    lims(x = c(30, 60),
         y = c(12, 23))
```

**Don't forget to add a code chunk, before this one, load your packages!**

**Question 1**

**part a**

Extract a data frame from `penguins` that excludes the Adelie penguins *without using* a data pipeline. Copy the code you used here.

**part b**

Now, *use a data pipeline* to extract a data frame that excludes the Adelie penguins. Save it into a new object! Copy the code you used here.

**part c**

Continue the pipeline by piping the data frame you just created to `ggplot()` to create a scatter plot of bill length vs. bill depth. You'll see that you can do data transformations in a pipeline leading up to a single plot, if you don't need to keep the transformed dataframe around once the plot is done. Copy your extended pipeline here.

**Question 2**

In a single pipeline, extract a data frame that 1) excludes the Adelie penguins 2) retains only penguins with bill lengths between 40 and 50 mm. Save the result into a new variable called `noadelie_medium`. Copy the code you used here.

**Question 3**

Sort the data frame from **Question 2** in decreasing order by bill length. Copy the code you used here.

**Question 4**

In a single pipeline, calculate the mean bill length and bill depth *for each of the three species of penguins*. Copy the code you used here. Do those statistics line up with what you see in the penguins plot provided by the (original) code given?

**Question 5**

Consider a new metric called `bill_size` that's the sum of the length and depth. In a single pipeline, 1) create this this new `bill_size` column as the sum of bill length and depth, 2) calculate the average bill size and its standard deviation among each species-island combination, and 3) sort your resulting data structure in decreasing order by average bill size. Copy the code you used here.

**Question 6**

Write a single pipeline to calculate the total number of penguins belonging to each species-island combination. Copy your code here:

**Question 7**

Write a single pipeline to calculate the proportion of penguins with body mass greater than 4000 grams in each species. Copy your code here: